Denisas Bykovas

System Architect Technical Lead Senior .NET Engineer

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PROFILE

System architect and hands-on technical lead with 25 years in .NET, most of them on systems that are regulated, national-scale or both: Lithuania's tax filing platform (EDS4, 10M+ filings a year), the Bank of Lithuania's SEPA gateway (CENTROlink, 757k+ payments a day), open-banking and product-data platforms for a Lithuanian bank, and a manufacturing execution system that has run a factory floor since 2021. .NET Foundation member since 2020.

I draw the architecture and stay close enough to the running system that the decisions still hold after the deploy: backend, integration, data, DevOps, and enough front-end to ship. I have built an engineering office from zero to 12 and led delivery teams of up to 20, and I still commit code every week. In live systems I prefer surgical changes to rewrites, treat legacy and regulation as constraints rather than obstacles, and leave each team with less debt than I found. In 2026 I also designed, built and shipped DEBYKO, a crypto market-data platform with its own MCP server, end to end as a solo founder.

Open to Senior / Principal Architect, Technical Lead, Head of Engineering and Senior .NET Engineer roles in the EU, on-site or remote. Available from January 2027.

EXPERIENCE

Technical Lead, Banking & Fintech — UAB Baltic Amadeus, Vilnius Jan 2026 – Jan 2027

Client: a regulated Lithuanian bank (under NDA).

Technical lead across three delivery streams — an open-banking payment gateway, a greenfield product-data platform and API-gateway governance — on .NET 9/10, Azure and Kubernetes, working directly with the bank's business and architecture teams in Scrum.

- Open Banking (PSD2) payment gateway: led backend delivery of OBIE and Berlin Group NextGenPSD2 bank connectivity — mTLS, OAuth2 client credentials, JWS request signing, consent and payment-status polling. ASP.NET Core, EF Core, PostgreSQL, Helm, six environments.
- Ran a 12-task performance programme: partial and trigram indexes, deterministic pagination, streamed CSV export, single-pass authentication, bounded-channel consumers, a token cache with 401/403 eviction. Removed ~78 ms of CPU per request that had capped a 1-core pod at ~13 rps.
- Implemented X.509 chain validation across several CA hierarchies (QWAC/QSeal, OCSP/CRL) so the planned CA migration needs no code change; enforced Clean Architecture across ~250 files; added a fail-fast schema-drift guard and a feature-flagged, self-migrating API-key hashing upgrade.
- Product Catalog platform (greenfield): architected and delivered eight .NET 10 microservices (FastEndpoints, EF Core 9, Azure SQL, AKS/ArgoCD, five environments) — three source-system connectors (Oracle ODS, MS SQL delta tables, REST) with watermark-based incremental sync and SqlBulkCopy/MERGE upserts, and an event-driven exporter publishing customer-product snapshots to Azure Service Bus with SHA-256 delta detection, sized for 500k customers in a 30-minute window.

.NET 9/10 · ASP.NET Core · EF Core · FastEndpoints · PostgreSQL · Azure SQL · Oracle · Azure Service Bus · AKS · Helm · ArgoCD · Azure DevOps · APIM · PSD2 / OBIE / Berlin Group · mTLS · OAuth2 · JWS

NDA, Lead Architect, National & Enterprise Infrastructure — independent contracts 2016 – present

Engaged directly by public-sector bodies and regulated enterprises. Four of the systems below are national- or plant-scale and still running.

EDS4 — national tax declaration platform 2019 – 2022 (VMI engagements since 2016)

State Tax Inspectorate (VMI) · Lead Architect

Modular, zero-downtime replacement of Lithuania's tax filing platform, used by every taxpayer and accountant in the country: 10M+ filings a year, multi-agency integration, releases the regulator can audit on the wire, and tax-law changes that ship without a refactor cycle.

.NET · MS SQL · Oracle · event-driven · CQRS · Azure · OAuth2

CENTROlink — national SEPA payment gateway 2018 – 2022

Bank of Lithuania · Architect & Integration Lead

Lithuania's gateway to the pan-European payment rails, scaled and extended during its 2018 – 2022 expansion: SCT, SDD, SCT Inst and TARGET2; RT1, STEP2, TIPS and SWIFT connectivity; signed messaging (XAdES) over mTLS; MT/XML message formats; the DPS portal for commercial banks. 148 connected PSPs, 757k+ payments a day (2022 average), 99.6% availability. Deployed alongside the legacy processors — no flag day.

.NET microservices · SEPA · SWIFT · XAdES · mTLS

LKvitai.MES — shop-floor execution 2021 – present

Lauresta, window-blinds manufacturer · Architect, hands-on

Event-driven manufacturing execution system covering order intake, task dispatch, semi-finished labels, assembly, warehouse, accounting sync, delivery scheduling and installation tracking. 100+ daily operators; operator errors down 60%; tracking latency from minutes to seconds.

.NET 8 modular monolith · Blazor Server · Marten / PostgreSQL · MQTT · Node-RED · Vue kiosks · Traefik · Docker

NDA water-refill infrastructure 2020 – 2021

Vilniaus Vandenys · Architect

City-wide embedded system for 16 unmanned drinking-water stations across Vilnius: edge computer vision for vehicle recognition, real-time telemetry, control logic at the hardware, aggregation and access control on the cloud side.

Embedded Linux · C++ / .NET · computer vision · MQTT

NDA global insurance 2021

.NET engineer, contract

.NET 6/8 microservices integrated with SharePoint Online for document storage and approval workflows; SharePoint Server 2016/2019 to Microsoft 365 migration (Sharegate, SPFx, Power Automate).

Other engagements: platform-engineering contract in the finance sector (Reiz Tech, 2025) — Chainguard hardened container images in GitHub Actions CI/CD, Red Hat OpenShift; subcontracted deliveries for public-sector procurements (Codigi, 2025 – 2026).

Founder — DEBYKO (MB Debyko), Vilnius 2026 – present

Crypto market-data platform designed, built and shipped solo in 2026: perpetual-futures and spot data from 20+ venues (Binance, Bybit, OKX, Hyperliquid, Kraken Futures, dYdX, GMX and others), stored exactly as each venue published it, with the venue's timestamp and the receive time on every value; stale values are flagged with a reason rather than shown as current.

- Products: Studio (cross-venue comparison and charts), REST API with point-in-time history and a cross-venue query language (DQL), Agent (DQL conditions that fire to Telegram or webhooks), Arena (a trading-bot scoreboard verified against the public tape).
- A daily Merkle root of all rows is anchored on Base, so any row can be verified independently.
- Remote MCP server (mcp.debyko.com) listed in the Anthropic, OpenAI, Glama and Smithery catalogues — the data is queryable from Claude, ChatGPT and Cursor in plain language.

.NET · PostgreSQL · venue WebSocket/REST feeds · MCP · Base · self-hosted infrastructure

Managing Director & Engineering Manager — Flinke Folk AS, Lithuanian branch 2014 – 2017

Built the Vilnius engineering office from zero to 22 developers for a Norwegian software house serving oil & gas clients. Architected and delivered the logistics, vessel-routing, materials-delivery, HR-planning and project-cost-estimation tools used in Shell's onshore operations in Norway, Malaysia and the USA. Hiring, engineering standards, CI/CD and delivery across time zones.

.NET · MS SQL · Azure AD · Agile

Software Developer & Systems Administrator 2004 – 2014

Lauresta, Galeo, Euroliuksas, freelance

- Lauresta (manufacturer, Klaipeda): full-stack ownership of the in-house ERP — 70+ applications and services (C#/VB.NET, ASP.NET MVC, PHP, MS SQL, Reporting Services) — and of the whole infrastructure: Windows Server 2003–2012, Exchange, VMware, ISA/Cisco, VPN to foreign branches, Crouzet PLCs on the production line.

- Galeo / Luktarna (2012 – 2014): distributed inventory and payment-card system for a 400+ gas-station network — C agents on the cash registers, C#/WPF core, Java mobile UI, ASP.NET MVC self-service, MS SQL, 24/7 reconciliation.

Earlier (1999 – 2003): accounting and retail-operations management. Writing software since 2000.

SKILLS

Architecture — system and solution architecture for regulated, high-availability platforms; microservices and modular monoliths; event-driven design, CQRS, DDD, Clean Architecture; C4, BPMN, UML; TOGAF; zero-downtime rollout and legacy co-existence; zero-trust, OWASP, threat modelling.

.NET and backend — C#, .NET Framework through .NET 10, ASP.NET Core, EF Core, Dapper, Marten, FastEndpoints, gRPC, Blazor; HybridCache, bounded channels, resilience pipelines; xUnit, NUnit, architecture tests.

Data — MS SQL Server, Oracle (PL/SQL), PostgreSQL (partial and trigram indexes, event store), Azure SQL, Redis, InfluxDB; SqlBulkCopy/MERGE pipelines, DbUp and EF migrations; Kafka, Azure Service Bus, MQTT.

Payments, banking and regulatory — SEPA (SCT, SDD, SCT Inst, TARGET2), RT1, STEP2, TIPS, SWIFT; PSD2 Open Banking (OBIE, Berlin Group NextGenPSD2); eIDAS QWAC/QSeal, XAdES, mTLS, OAuth2/OIDC, JWT, Entra ID; national tax e-services.

Cloud and platform — Azure (AKS, API Management, Service Bus, Functions, Key Vault, Managed Identity, Logic Apps), Kubernetes, Helm, ArgoCD, Docker, Azure DevOps, GitHub Actions, APIOps/GitOps, Red Hat OpenShift, Chainguard images, Terraform and Bicep basics; Proxmox, Traefik, Caddy, WireGuard; Prometheus, Grafana, Serilog.

Front-end — ASP.NET Core MVC, Blazor Server, React, Vue, WPF; SharePoint Online and SPFx; Power Apps, Power Automate.

AI-assisted engineering — author of a production MCP server (DEBYKO); daily work with Claude Code, OpenAI Codex, Cursor and GitHub Copilot; spec-first workflow (architecture blueprint → agent implementation → review); LLM tool-calling and agent rules in shipped features.

Embedded and IoT — embedded Linux, C/C++, Raspberry Pi, microcontrollers and VHDL, PLCs, edge computer vision.

EDUCATION AND RESEARCH

- **Doctoral studies, Informatics Engineering** — Vilnius University, 2014 – 2019. Research on semantic model-based interoperability of next-generation enterprise information systems; dissertation not defended.
- **MSc, Technical Information Systems Engineering** — Klaipeda University, 2012 – 2014 (GPA 9.6/10); Erasmus term in Robot Systems at the University of Southern Denmark, 2014.
- **BSc, Informatics Engineering** — Klaipeda University, 2009 – 2012, the five-year programme completed in three.

Publications

- *Self-learning adaptive algorithm for maritime traffic abnormal movement detection based on virtual pheromone method* — IEEE SPECTS, 2015.
- *Adaptive artificial neural network-based control of wind turbines* — Technologijos mokslo darbai Vakaru Lietuvoje, 2014.

CERTIFICATIONS AND MEMBERSHIPS

- Microsoft Certified: Azure Administrator Associate — 2024
- MCSD: App Builder — 2021 · MCSA: Web Applications — 2019
- .NET Foundation member since 2020

OPEN SOURCE AND PERSONAL PROJECTS

- **DEBYKO MCP server** — public remote MCP server for crypto market data; listed in the Anthropic, OpenAI, Glama and Smithery directories.
- **BYKO** — immutable ERC-20 on Base (Solidity, OpenZeppelin, verified contract) with an Aerodrome liquidity pool; site and brand built from scratch.
- **ShopfloorPilot demo** — Rete.js v2 / Vue flow editor that turns production rules into executable task graphs.
- **AquaHub, reefconsole** — .NET and Python reef-aquarium automation on Raspberry Pi: lighting, auto top-off, pH and temperature telemetry into InfluxDB and Grafana.

github.com/bykovas

LANGUAGES

Lithuanian — native · Russian — native · English — fluent written, professional spoken · Ukrainian — basic

INTERESTS

Reef aquariums and the automation around them · 3D printing · self-hosted infrastructure · digital photography and video · computer vision and neural networks